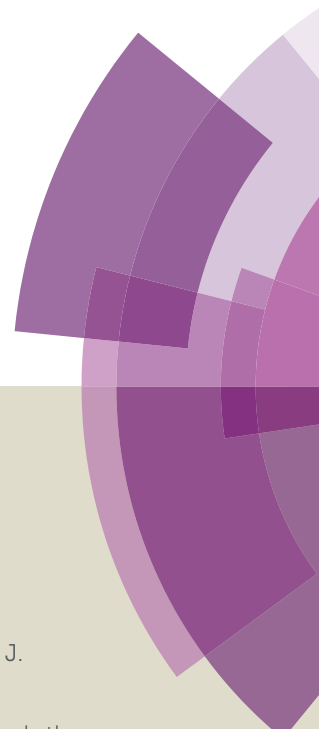


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ARTICLE

Ultrathin Paper-like Boron-doped Carbon Nanosheet Electrodes Combined with Boron-enriched Gel Polymer Electrolyte for High-Performance Energy Storage

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Flexible energy storage devices that can function under considerable physical deformation have shown great promise in the applications of portable electronics. In this work, we report for the first time a novel design of all-solid-state symmetric supercapacitors (SCs) based on free-standing ultrathin Boron-doped graphene paper (B-GP) electrodes with Boron-enriched gel polymer electrolyte (GPE). Specifically, the B-GP electrodes integrated with B-contained GPE not only introduce more electrochemically active sites for absorption/desorption of electrolyte ions, but also facilitate diffusion of electrolyte during charging/discharging processes. The novel design of B-GP with B-contained GPE endows our symmetric SC a voltage of 2.5 V, an unprecedented energy density of 39.31 Wh kg⁻¹ (with power density of 1.44 kW kg⁻¹), a stable cycling performance (capacitance retention of 90% after 3000 continuous charge/discharge cycles), good rate capability, distinguished mechanical flexibility and temperature resistant stability. This work provides a promising candidate to design new generation of electrochemical SCs for energy storage devices.

Introduction

Nowadays, with increasing requirements of modern gadgets, energy storage devices such as supercapacitors, Li-ion batteries and fuel cells are a subject of immense interest.¹⁻⁶ Conventional commercial devices like SC, which is mainly made up of rigid electrodes, cannot be bent or twisted. Thus, to meet the demand of practical applications ranging from flexible/bendable electronic equipment to microelectro-mechanical systems, significant research has been focused on the development of flexible and lightweight energy storage devices over the past few years.^{7, 8} Flexible SCs have attracted significant attention for its large power density, moderate energy density, distinguished mechanical flexibility, long cycling life and good operational safety.⁹⁻¹¹ Therefore, great efforts have been devoted to design new electrodes materials and polymer electrolytes, which combine outstanding electrochemical properties with promising mechanical flexibility.¹²⁻¹⁴

Meanwhile, nickel foams or metal foils are used as current collectors in traditional two-electrode 'sandwich' configuration. In addition, conductive additives and binders (such as poly(vinylidene fluoride) or polytetrafluoroethylene) besides the active electrode materials in the electrodes, bring apparent

parasitic weight to the cell assembly that leads to an undesirably low electrode material to collector/separator ratio, decreased conductivity, and possibly a low electrochemical columbic efficiency.^{9, 15-17} Thus, it is of great significance to design and fabricate novel electrode materials for SC devices assembly that are ultralight weight and mechanically robust without additives or current collector, which in turn further increases the gravimetric capacitance and energy density of the packaged SCs.

Carbon materials have already shown various applications in electrodes, catalysts, and supports for energy devices such as supercapacitors, batteries, fuel cells, and hydrogen storage materials.^{3, 18-20} Moreover, recent studies showed that graphene-based paper-like structures present unique electronic, mechanical and chemical properties such as high electrical conductivity, flexibility, high mechanical strength and chemical stability.²¹⁻²³ The paper-like structure is suggested to be the result of the interfacial activity of graphene oxide sheets, and free-standing graphene paper (GP), has already been prepared by vacuum filtering, self-assembly and pressing graphene aerogel or hydrogel.²⁴ Based on the above advantages, GP is prove to be an alternative approach to fabricate a flexible SC.^{13, 25-27}

However, a flexible SC assembled with GP is still not qualified for practical applications, for only electrical double-layer capacitance is available.^{28, 29} To improve the performance of GP, chemical doping has been reported to be one of the most effective methods in tailoring the electronic properties of graphene-based materials, and heteroatom doped GP has recently attracted enormous interests.³⁰ The dopant atom can generate an activation region on the graphene surface by influencing the spin density and the charge distribution of

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neighboring carbon atoms, providing additional pseudocapacitance at the interface and promotes the performance.^{31, 32} N-, S- or B-doped graphene-based materials have shown enhanced electrochemical properties. For example, Zhang et al. reported the amorphous phosphorus/nitrogen-doped GP for sodium-ion batteries anode materials, which exhibits an ultrastable cyclic performance and excellent rate capability (809 mA h g⁻¹ at 1500 mA g⁻¹).³³ Han et al. demonstrated that a free-standing N-doped GP can be used as binder-free electrodes for the lithium/dissolved polysulfide batteries, which exhibited high specific capacities of approximately 1000 mA h g⁻¹ after 100 cycles.³⁴ Akhter et al. fabricated a novel flexible three-dimensional (3D) architecture of nitrogen and sulfur codoped graphene, and it displays outstanding capacitive performance of 305 F g⁻¹ (at 100 mV s⁻¹) and energy density of 28.44 Wh kg⁻¹ (at the power density of 930 W kg⁻¹).³⁵ Pure B-doping is reported to exhibit a remarkable supercapacitor performance which is close to that of boron- and nitrogen-codoped sample.³⁶ The boron atom has one electron less than carbon in the valence layer and lower electronegativity, which induces changes on both the conductive properties and the oxidation behaviour of the carbon materials.^{37, 38} It has also been found that B-doping in other types of carbon materials remarkably promoted the performance when used as SC electrodes.³⁹⁻⁴¹ However, to the best of our knowledge, there has been no investigation on the preparation of B-doped GP and its application in supercapacitors.

Polymer electrolytes for supercapacitors with borate or borane moieties have more promising future over traditional polymer electrolytes due to its effects on increasing ionic conductivity through electron withdrawing group oxaborinane capturing anions from electrolyte salts. Several works have been conducted to study this kind of polymer electrolytes.⁴²⁻⁴⁴ Li et al. reported electrolytes using tris(pentafluorophenyl) borane (TPFPB) or THFPB as anion receptor presenting the ionic conductivity of 30 mS/cm at 20 °C in 0.6 M LiF-EC:DMC (1:1).⁴⁴ Recently, our group have successfully synthesized a novel B-contained flexible polymer electrolytes, poly((2-phenyl-1, 3, 2-dioxaborolan-4-yl) methyl methacrylate) (P(GMMA-PBA)), in which the heterocyclic borane groups of polymer side chains are good anion receptors and exhibits good electrochemical stability.⁴⁵

Herein, the free-standing B-doped GP, as well as the integration of B-doped GP as the electrodes and B-contained GPE as the polymer electrolytes for SC devices, are both reported for the first time. The flexible and lightweight symmetric SC exhibits intriguing electrical properties, giving maximum energy density of 39.31 Wh kg⁻¹ (at a power density of 1.44 kW kg⁻¹) and maximum power density of 16.81 kW kg⁻¹ (at an energy density of 27.55 Wh kg⁻¹). This study opens a new approach to develop high-performance flexible energy-related devices and wearable electronics.

Experimental

Materials preparation

Glycerylmono-methacrylate (GMMA, J&K Scientific 95%) was passed through a basic alumina column to remove the inhibitor. Phenylboronic acid (PBA, Purum, ≥97.0%, Aldrich) was used as received. Lithium perchlorate (LiClO₄, battery grade, dry, 99.99%, Aldrich) were dried under vacuum at 130 °C for 48h prior to use and stored in an argon filled glove box (MBRAUN, LABstar, H₂O < 1 ppm, O₂ < 1 ppm). Benzoyl peroxide (BPO, Aldrich) was washed with chloroform (CH₃Cl, Sinopharm Chemical Reagent Co., Ltd, 99.9%) and precipitated by methanol (Sinopharm, 99.9%) and then stored in refrigerator. Dichloromethane (DCM, Sinopharm, 99.9%) was refluxed over CaH₂ before use and stored with molecular sieves. Dimethylformamide (DMF, Sinopharm, 99.9%) was distilled under reduced pressure and stored with molecular sieves. Molecular sieves (4Å, Sinopharm) and CaH₂ (Sinopharm) were used as received. Lithium borohydride (LiBH₄, Aldrich, 95%), Tetrahydrofuran (THF, Aldrich, 99.9%), GO (Tianjin Plannano Co., Ltd.) were obtained commercially. LiBH₄ was ball-milled for 2 h before used. Tetrahydrofuran was kept over CaH₂ and distilled at 339 K (66 °C).

Synthesis of Boron-doped Graphene Paper (B-GP)

GO (0.1 g) was dispersed in anhydrous tetrahydrofuran (30 ml). Ball-milled lithium borohydride (0.1 g) was dissolved in 20 ml of anhydrous tetrahydrofuran with mild sonication for 0.5 h. The two solutions were then mixed together under N₂ atmosphere and refluxed (reaction temperature, 95 °C) for 3 days. Finally, the B-doped graphene was produced after filtration, washing with methanol and deionized water for several times, and drying at 60 °C for 24 h. The thickness and mass loading of as-filtered graphene paper can be controlled simply by adjusting the volume of well-dispersed solution. The as obtained graphene paper was named B-GP-1. In similar way, other samples like B-GP-3, B-GP-5 were prepared by varying the concentration of lithium borohydride viz. 0.3 g, 0.5 g respectively.

Preparation of Gel Polymer Electrolyte film

The synthesis of monomer GMMA-PBA was referred according to our previous report.⁴⁵ Gel polymer electrolyte membrane containing heterocyclic boron moieties was prepared by one-pot thermal polymerizing technique. 2 ml monomer GMMA-PBA with 1 ml 1 M electrolyte salts (DMF/LiClO₄) were mixed together. After 10 wt% BPO dissolved in the solutions completely, the mixtures were pour into a 0.5 mm thick PTFE template to polymerize in a dry oven for 6 h at 85 °C. All the preparation of the GPE membrane were conducted under inert gas protected conditions.

Characterization

The morphology of the as-prepared samples were examined by field-emission scanning electron microscope (FESEM, FEI Nova NanoSEM 450), transmission electron microscope (TEM, FEI Tecnai G2F20S-Twin) and Atomic force microscopy (AFM,

Bruker Dimension Icon). Raman spectra were recorded on a Dilor LABRAM-1B multi-channel confocal microspectrometer with 514 nm laser excitation. The surface species were detected by X-ray photoelectron spectroscopy (XPS, Kratos Axis Ultra) with a monochromatized Al K α X-ray source as the excitation source ($h\nu = 1486.6$ eV). The experimental uncertainty of the XPS binding energy was ± 0.1 eV. FT-IR spectra were obtained on a Fourier transform infrared (Nicolet Nexus 470) spectrometer in the range from 4000 to 400 cm^{-1} . Thermogravimetric Analysis (TGA, TA Instruments Q500) was conducted to determine the composition of the electrode B-GP and GPE with a ramp rate of 10 $^{\circ}\text{C min}^{-1}$ under N_2 atmosphere.

Fabrication of symmetric flexible solid-state supercapacitors

All electrochemical measurements were performed with a CHI 660E electrochemical workstation (Shanghai CH Instruments Co., China) in a two-electrode capacitor configuration. In the case of all-solid-state SCs, the B-GP were cut into rectangular strips and were directly used as working electrodes without any other polymer binders or conductive additives. The thickness of each electrodes was controlled to be around 15–25 μm , and the corresponding areal mass of B-GP was 1.4 mg cm^{-2} . The electrode were immersed into 1 M $\text{LiClO}_4/\text{DMF}$ solution overnight prior for the device assembly. The as-synthesized GPE membrane was sandwiched between two same electrodes and transferred into a PET bag. Then the PET bag was laminated using a roller press under a pressure of 1 MPa for 10 min and encapsulated. The assembly process was conducted in a glove box.

The specific capacitance of per single electrode in the symmetric cells can be obtained as following (1):

$$C_s = \frac{I \times \Delta t}{M \times \Delta V} \quad (1)$$

where C_s is the specific capacitance of a symmetric supercapacitor, I is the discharge current in Ampere (A), Δt is the discharge time (s), ΔV is the operating voltage window obtained from the discharge curve excluding the voltage drop, M is the total mass (g) of active materials in the symmetric supercapacitor.

The energy density and power density of symmetric supercapacitors are calculated using the following equation (2–3):

$$E = \frac{1}{2} C_s V^2 \quad (2)$$

$$P = \frac{E}{t} \quad (3)$$

where V is the actual voltage excluding IR drop of the discharge process, t is the discharge time (s).

Results and discussion

The free-standing and flexible B-doped graphene papers were prepared through the reduction of graphene oxide in different concentration of lithium borohydride-THF solution (see Fig. 1

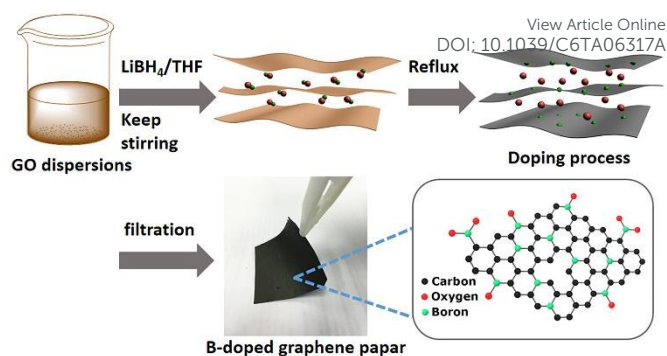


Fig. 1 Schematic demonstrating the concept of self-assembling the boron-doped graphene paper.

and $\text{S1}^\dagger$). The products were denoted as B-GP- x , with x representing the mass ratio of lithium borohydrides to graphene oxide.

Field emission scanning electron microscope (FE-SEM) is used to characterize the as-prepared B-GP- x ($x=1, 3, 5$) samples (Fig. $\text{S2}^\dagger$). The SEM images of B-GP-1 and B-GP-3 reveal wrinkle-like morphology on its surface, which may be due to defects from the breaking of sp^2 bonds in the original graphite.^{46, 47} With increasing the amount of LiBH_4 , the B-GP-3 paper shows more wrinkles than B-GP-1, indicating higher degree of reduction and doping. Thus, the B-GP-3 exhibits better electrochemical performance than B-GP-1. However, large amount of borates particles are aggregated on the graphene sheets of B-GP-5, which hinder the contact of the electrolyte with active materials and the charge transfer during charging/discharging process, thus decreasing its comprehensive properties.^{40, 48} The XRD spectrum of B-doped graphene papers have also proven the SEM results (Fig. $\text{S3}^\dagger$).

Fig. 2a shows the optical image of B-GP-3 paper. The as-prepared smooth GP exhibits excellent mechanical flexibility, which allows it to be rolled up, twisted or bent (Fig. $\text{S1}^\dagger$). The B-doped GP features a transparent sheet structure in the high resolution transmission electron microscopy (HRTEM) image (Fig. 2b). The HRTEM image reveals that the B-doped GP is ultrathin, which is also verified by atomic force microscopy (AFM) measurement to be about 1.8 nm. Additionally, there are many corrugations and scrollings on the surfaces and edges of transparent graphene sheets, which may improve the rate capability of supercapacitor (Fig. $\text{S4}^\dagger$).⁴⁹ The B-doped graphene sheets are microscale flat in the lateral dimension and only few layers thick with wrinkles. Richly-wrinkled structure is favorable for the diffusion of electrolyte into the inner of the B-GP electrodes.^{50, 51} The elemental composition is revealed by the energy dispersive X-ray spectroscopy (EDS) studies (Fig. $\text{S4}^\dagger$). The EDS images revealed that the element B is homogeneous distributed on the graphene surface, as observed in SEM image. Cross-sectional SEM images of the B-GP-3 paper shows well-layered structure with a uniform thickness of about 20 μm (Fig. 2c and d). Owing to the hydrogen release through the decomposition of LiBH_4 , the loose and porous structure of B-GP-3 is formed, which is expected to facilitate electrolyte ions transport between the electrode surface and electrolyte.

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Fourier transform infrared spectroscopy (FT-IR) spectra of GO and B-GP-3 are shown in Fig. 3a and b (for B-GP-1 and B-GP-5, Fig. S5[†]). The broad and intense peak of O-H groups centered at 3433 cm⁻¹ and the C-O stretching peak centered at 1057 cm⁻¹. The peak at 1728 cm⁻¹ was the characteristic of C=O stretch of carboxyl groups situated at the edges of the GO sheets. In the spectrum of the as-prepared B-GP-3, all the peaks were significantly diminished in intensity, which suggests that a large portion of the hydroxyl and epoxy groups on the basal plane is removed by the lithium borohydrides-THF.

Raman spectroscopy is used to measure the density of defects in graphene-based materials. In a Raman spectrum, the D band at approximately 1350 cm⁻¹ corresponds to structural defects and disorders in the carbon matrix, and the G band at approximately 1570 cm⁻¹ is related to the E_{2g} vibration mode of sp²-bonded carbon atoms.⁵² The intensity ratios of I_D/I_G were 0.98 for GO and 1.17 for B-GP-3, respectively (Fig. 3c). The Raman spectra of B-GP-1 and B-GP-5 are given in Fig. S5.[†] These results indicate that the increase in the I_D/I_G values can

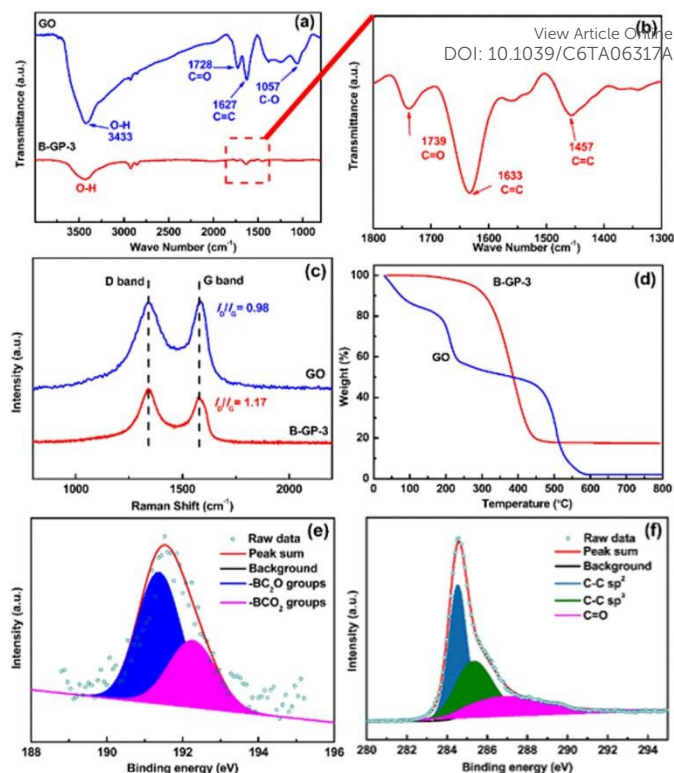


Fig. 3 (a-b) FTIR spectra, (c) Raman spectra, and (d) TGA of GO and B-GP-3 paper. (e) The high-resolution B1s peaks, and (f) high-resolution C1s peaks of the B-GP-3 paper.

be attributed to the higher densities of structural defects and disorder consequent of doping.

As illustrated in Fig. 3d, thermogravimetric analysis (TGA) curves show that B-GP-3 is more thermally stable than GO. GO showed significant weight loss below 100 °C due to emission of water. The second apparent weight loss stage of GO processed in the temperature range of 180–220 °C, which is attributed to the decomposition of oxygen-containing groups.⁵³ Meanwhile, the weight loss of B-GP-3 (3.3 wt %) at 220 °C is significantly less than that of GO (37.1 wt %). There are almost no residual at 600 °C for GO, while the residue of the B-GP-3 is 17.6 wt %.

The X-ray photoelectron spectroscopy (XPS) is also a useful analysis technique that can provide information of the material. The C/O ratio is used to evaluate the degree of reduction in B-GP. A typical C/O ratio of GO is about 2.34, while it increases to 5.14 for B-GP-3 after reduction by LiBH₄ (Fig. S5[†] and Table S1[†]). Deconvolution of the B1s spectra produces two main peaks around 191.6 and 192.4 eV (Fig. 3e). They have been assigned to boron species with mixed B-C and B-O bonds, mainly the –BC₂O groups and the –BCO₂ groups.⁴⁰ Moreover, since these BCO moieties (–BC₂O and –BCO₂) are unstable within the graphene lattice, these moieties may be preferentially located at the chemically active edges.⁵⁴ The high-resolution XPS core level spectra of C1s spectra is well fitted by three peaks (Fig. 3f). These peaks are assigned to C=C sp² delocalized bonds (284.5 eV), C-C sp³ bonds (285.3 eV) and C=O carbonyl bonds (286.8 eV), respectively.⁵⁵ As described above, these chemical analyses evidently show the successful reduction of graphene oxide and the resulting B-GP-3 material has B-doped carbon networks.

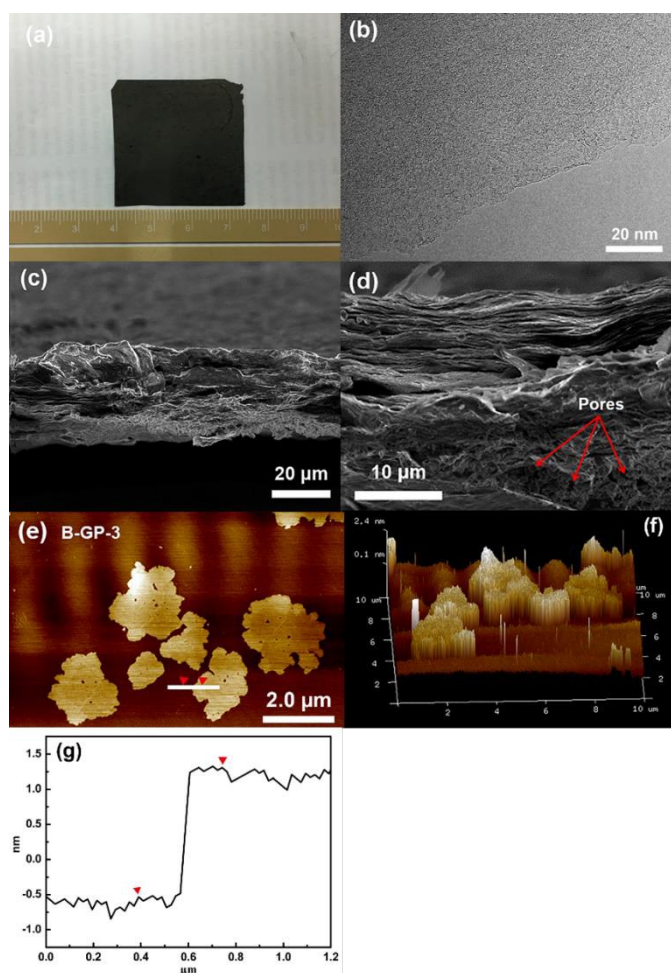


Fig. 2 (a) Digital photograph of the large-area B-GP-3 paper. (b) HR-TEM image of the B-GP-3 paper. (c-d) The SEM images of the fractured cross sections with low and high magnifications. (e) Typical tapping mode AFM images, (f) 3D AFM images and (g) the height cross-sectional profiles of B-GP-3.

The free-standing B-GP were directly used as electrodes to fabricate all-solid-state symmetric supercapacitors with B-contained GPE. B-contained GPE membrane (P(GMMA-PBA)) was synthesized as our previous report (Fig. S6[†]). Cyclic voltammetry (CV) tests were firstly conducted to obtain the stable potential window of the assembled SCs with a scan rate of 50 mV s⁻¹ as shown in Fig. 4a. The stable voltage window of the SCs can achieve 2.5 V. CV measurements at various scan rates are carried out and shown in Fig. 4b. Obviously, the CV curves of B-GP display a quasirectangular shape, suggesting that the as-prepared B-GP exhibits the ideal capacitance behavior and fast charging/discharging process characteristics.⁵⁶ A pair of anodic and cathodic wide humps is observed, which is an indication of the presence of reversible faradic reactions that are associated with electrochemically active functionalities on the electrode surface. The rectangular-like shape indicates typical double-layer capacitance, while a pair of redox humps in the CV curves reflect the pseudocapacitive contribution.^{13, 57, 58} It is evident that the current level of B-GP-3 are larger than that of the others at the same scan rate indicating the superior electrochemical performance and higher specific capacitance. It can also indicate a low resistance of B-GP to mass transfer and good charge propagation of ions at the interfaces between the GPE film and the B-GP electrodes. As shown in the CV measurements, the current level of B-GP-3 are obviously larger than that of B-GP-1 at the same scan rate, indicating the superior electrochemical performance and higher specific capacitance of the electrode material with the increased boron-doping amount. Fig. 4c displays the galvanostatic charge/discharge (GCD) curves of the SCs at different current densities. The GCD curves of B-GP electrode show a symmetric behavior with slight deviation of curvature, indicating that the supercapacitive behavior is influenced by both the electric double layer and pseudocapacitance as in accordance with the CV curves.⁵⁹ The linear potential variation with time attributes to the double layer capacitor behavior, while the nonlinear potential variation signifies the pseudocapacitor behavior.⁶⁰ Comparison of GCD tests with various current densities for as-prepared B-GP-1, B-GP-3 and B-GP-5 electrodes are shown in Fig. S7.[†] Fig. 4d displays the capacitance of the SCs as a function of current density. The highest C_s for the flexible SC device with P(GMMA-PBA) electrolyte is 47.15 F g⁻¹ at 1 A g⁻¹ on the basis of entire mass of two electrodes. The SC device also demonstrates a good rate capability with a capacitance retention of 70% as the current density increases from 1 to 10 A g⁻¹. This further confirms the B-doped graphene paper possess an outstanding electrochemical properties, which may come from both B- and/or O-functionalities of B-doped graphene paper.^{41, 61-64}

The all-solid-state flexible SC fabricated above also exhibits ideal capacitive performance at varied temperature ranged from 0 to 80 °C. As shown in Fig. 5a, the C_s remarkably increased from 38 F g⁻¹ at 0 °C to 105 F g⁻¹ at 80 °C. Moreover,

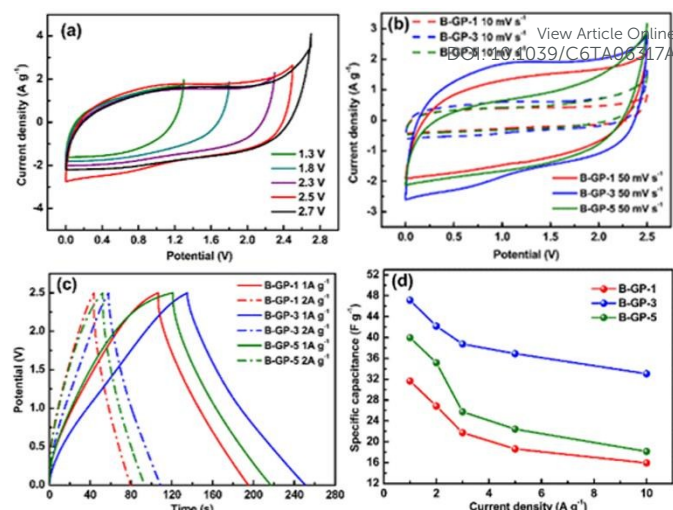


Fig. 4 (a) Cyclic voltammogram data for B-GP-3 electrode obtained with different upper potential limits. (b) Cyclic voltammogram data for three different B-GP electrodes over a range of scan rates. (c) Galvanostatic charge/discharge curves of three different B-GP electrodes over a range of current densities between 0–2.5 V. (d) Specific capacitance of B-GP-1, B-GP-3 and B-GP-5 papers as a function of discharge current density.

CV tests of the assembled SC devices obtained with different physical states of normal, bending, folding, and twisting at a scan rate of 50 mV s⁻¹ were shown in Fig. 5b. Apparently, all the CV curves still maintain quasirectangular shape, suggesting the reservation of ideal capacitance behavior and fast charging/discharging process characteristics regardless the physical state. Differing from the SCs packed with aqueous or organic electrolytes, the symmetric B-GP SCs using the GPE shows excellent mechanical stability (Fig. 5c).

The novel B-doped GP electrodes and B-contained GPE with excellent flexibility are suitable for highly flexible SCs (Fig. 6a). As the electrochemical stability is an important requirement for SCs application, the cycling performance of the SC device evaluated by consecutive charge/discharge cycles at a current density of 1 A g⁻¹ (Fig. 6c). Benefiting from the B-doped GP material and the B-contained gel electrolyte, our device exhibits outstanding cycling stability. After 3000 consecutive

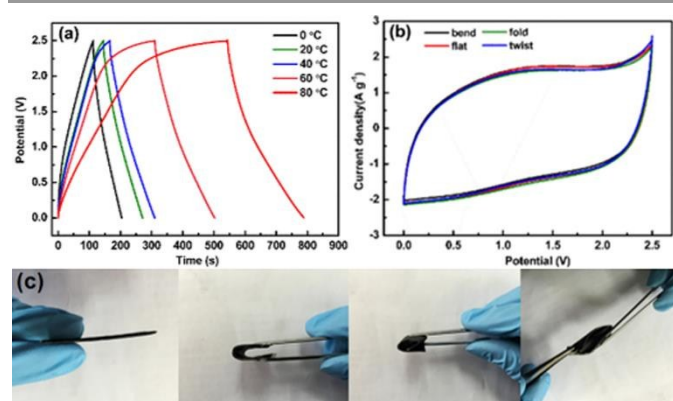


Fig. 5 (a) Galvanostatic charge/discharge curves of the as-assembled symmetric SC over a range of working temperatures at a current density of 1 A g⁻¹. (b) CV curves of the flexible SC device at different deformation states measured at a scan rate of 50 mV s⁻¹. (c) The assembled symmetrical capacitors exhibits highly flexible and mechanically stable properties with different physical states of normal, bending, folding, and twisting.

cycles, the Cs still remains 90% of its initial value. Cross-sectional view SEM images of the device before and after cycling are shown in Fig. 6b and 6c. It can be seen that after the consecutive cycles, the B-GP electrodes and the B-contained gel electrolyte still hold tight attachment, exhibiting excellent electrochemical stability of our flexible SC. Significantly, our device is able to lighten two red LED lamps up to 180 s after charging for only 20 s (inset in Fig. 6c and S8[†]), indicating its promising practical applications in energy storage. Electrochemical impedance spectroscopy (EIS) is also investigated to study the integral resistance of the SCs. As shown in Figure 6d, the Nyquist plots of our SC device are obtained by a frequency response analysis (FRA) of the frequency range from 0.1 Hz to 10^6 Hz at an amplitude of 5 mV. The intercept at high frequency on the real axis is the intrinsic internal resistance (R_s) of the electrode material and electrolyte of device. The semicircle gives the charge transfer resistance (R_{ct}), relative to the interface nature of electrolyte and electrode material.⁶⁵ The linear lines exhibit at low frequency region indicate the ideal capacitive behavior.⁶⁶ We notice that after the consecutive charge/discharge cycles at a current density of 1 A g^{-1} , the values of R_s are almost the same (from 12.39 to 13.68 Ω , Table S2[†]), which may be the result of the tight attachment between as-prepared ultrathin B-GP

electrodes and highly stable B-contained gel polymer electrolytes. It can be observed that the value of R_{ct} (0.27 Ω) is smaller than R_s , which may be due to the poor transfer of ions in gel polymer electrolytes. More comparison about R_s and R_{ct} in different electrolyte systems are shown in Table S3[†] and S4[†]. In addition, the nearly vertical lines at low frequencies suggest the low diffusive resistivity and quick motion of ions.⁶⁷ The measured impedance data are analyzed by fitting to an equivalent circuit (Figure S9[†]). And the electrochemical performance comparison between B-GP-1 and B-GP-3 is also presented (Figure S10[†]). Therefore, the B-GP electrodes combined with the B-contained gel electrolytes can be good candidate of high performance and outstanding charging/discharging rate stability materials for SCs. Ragone plot shows that our SC device also exhibits comparable power and energy densities among previously reported symmetric SCs using organic electrolyte (Figure 6e).^{68–72} The symmetric SC delivers a maximum energy density of 39.31 Wh kg^{-1} at a power density of 1.44 kW kg^{-1} and still maintains an energy density of 27.55 Wh kg^{-1} at a power density of 16.81 kW kg^{-1} .

Our results indicate that lithium borohydride (LiBH_4) can serve as an effective agent in both GO reducing and doping. Previous works have demonstrated that borane and ammonia borane can also act as reducing agent for GO and produce nitrogen- and/or boron-doped graphene materials with excellent supercapacitor performance.^{58, 73} Metal borohydrides, one of promising solid state hydrogen storage candidates, are investigated only for NaBH_4 .^{48, 74} LiBH_4 has caught enormous attention because of its light weight and therefore high gravimetric hydrogen content (18.51 mass %).⁷⁵ Recently, theoretical calculations demonstrating small clusters of LiBH_4 supported on graphene sheets have been performed by DFT method.⁷⁶ It is well known that the density of dangling bonds in amorphous carbons is large, about $2 \times 10^{20}\text{ cm}^{-3}$.⁷⁷ Thus, when single formula units of LiBH_4 are placed above the vacancy sites of the graphene sheet, boron finds a natural place in the lattice participating fully in the sp^2 bonding skeleton with a dramatic 6 eV reduction in energy. This finding explains well that why nanoparticle form of LiBH_4 decomposes easily with respect to bulk, for much of the boron becomes immediately trapped by strong bonding with unsaturated carbon.⁷⁶

Conclusions

We have successfully synthesized a novel, ultrathin boron-doped graphene paper via simple reaction of graphene oxide with lithium borohydride. Lithium borohydride is proved to be an effective graphene oxide reducing and doping agent, which introduce a large amount of B-C and B-O bonds into graphene. XPS analysis have demonstrated that B1s peaks were assigned to the $-\text{BC}_2\text{O}$ groups and the $-\text{BCO}_2$ groups, which could enhance electrochemical properties of materials by increasing pseudocapacitance. In addition, we have also fabricated for the first time a high-performance supercapacitor based on B-GP and B-contained GPE (P(GMMA-PBA)). The symmetric supercapacitors can reach a high specific capacitance of 47.15 F g^{-1} at room temperature, with only 10% degradation to the

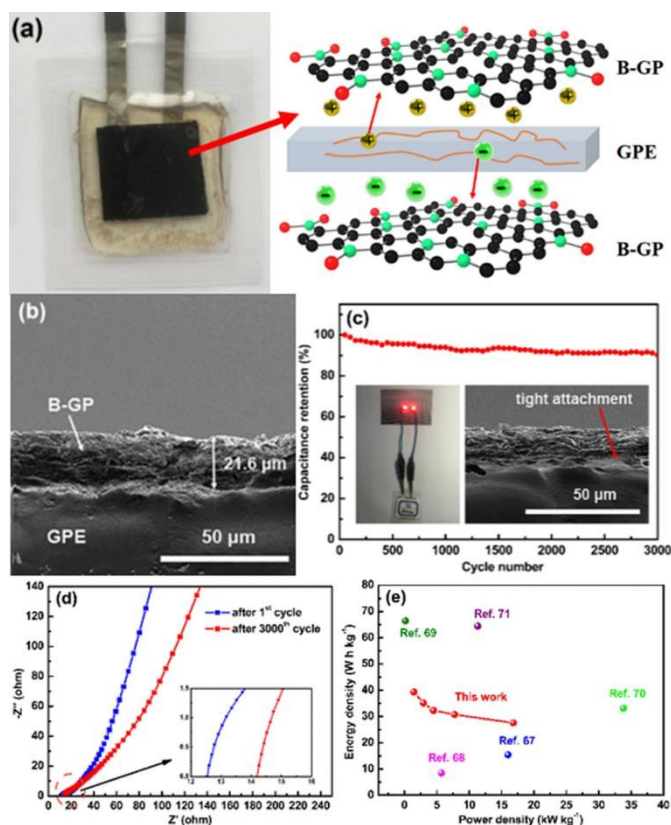


Fig. 6 (a) Schematic illustration of the structure of the symmetric SC. (b) The SEM image of the interface of B-GP electrode and GPE. (c) Cycling stability of the all-solid-state flexible SC at a current density of 2 A g^{-1} (inset left: photograph of two red LED lamps powered by device; inset right: SEM image of the interface after 3000 cycles). (d) Comparison of EIS spectra of the all-solid-state flexible SC after 1st and 3000th cycles. (e) Ragone plots of the flexible SC and others from the literature.

specific capacitance after 3000 cycles. The maximum energy density is 39.31 Wh kg⁻¹ (with power density of 1.44 kW kg⁻¹) and the maximum power density is 16.81 kW kg⁻¹ (with energy density of 27.55 Wh kg⁻¹). More importantly, given the highly flexible structure and close attachment, the as-fabricated SC devices exhibit excellent mechanical stability and outstanding tolerance at various temperatures. Further demonstration shows that the device can power up commercially available LED lamps. To the best of our knowledge, this work provides an unprecedented approach in preparing boron-doped GP and fabricating flexible energy storage devices as the power back-ups for stretchable/flexible electronic devices.

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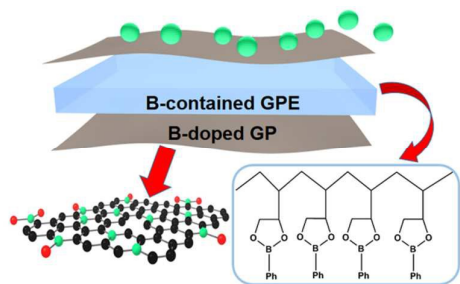
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Graphical Abstract



The novel design of B-doped graphene paper electrodes combined with B-enriched gel polymer electrolytes endows symmetric SCs outstanding electrochemical performances.